

DRAEM - ICCV 2021

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

Christopher Roßbach¹

¹Friedrich-Alexander-Universität Erlangen-Nürnberg, Technische Fakultät

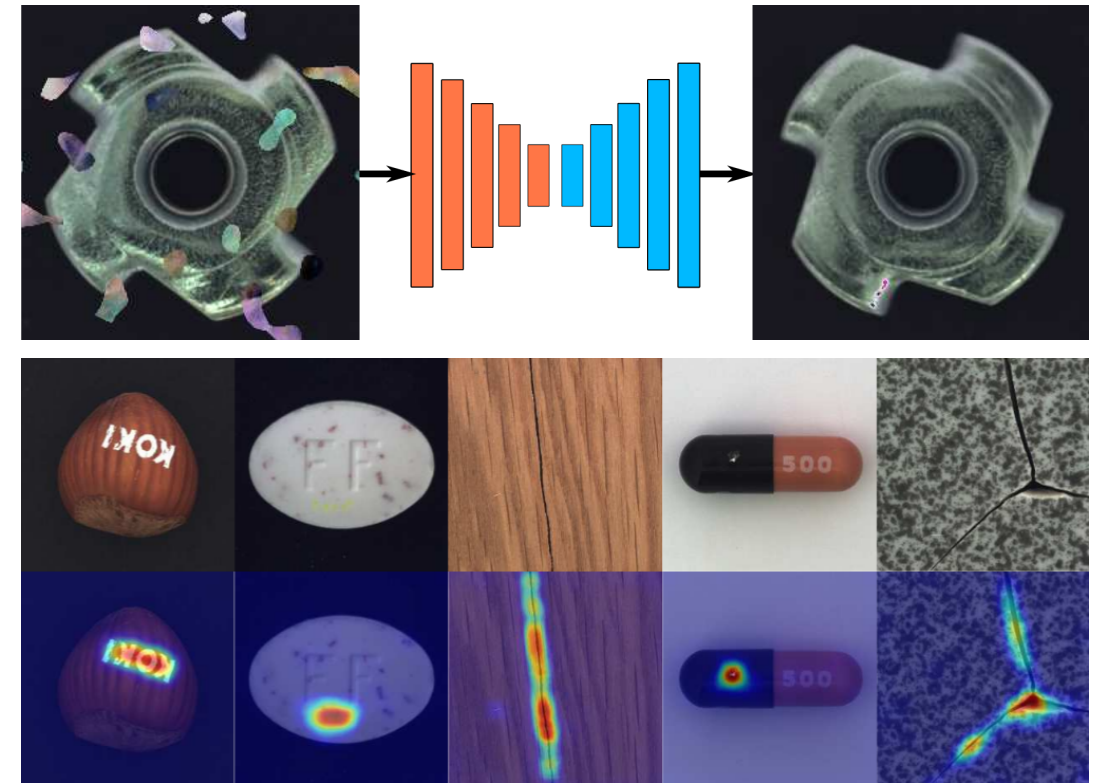
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- Detection and localization of anomalies in the image space, e.g. RGB, thermal, X-ray, ultrasound, ...
- Detecting skin lesions, defects in circuit boards, scratches on automotive parts, cracks in bridges, ...
- OOD-Samples are often rare and diverse
- Collecting data for all possible anomalies is often infeasible
- This gets worse when pixel-accurate localization is required
- Therefore, methods that only rely on normal data are preferred

Problem Statement

Current Situation and Related Work

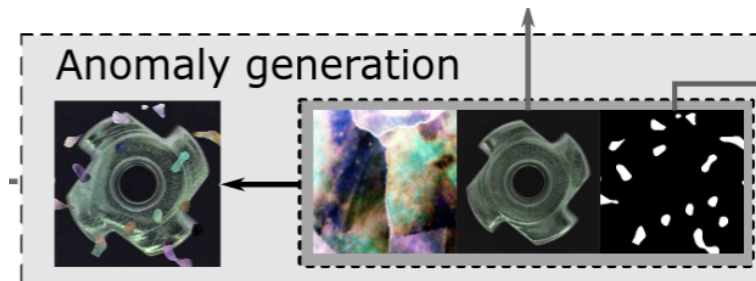
- **Paper:** “DRÆM – A Discriminatively Trained Reconstruction Anomaly Embedding for Surface Anomaly Detection”¹
 1. **Reconstructive Methods** (Autoencoders, GANs) detect anomalies by measuring reconstruction error
 - ▶ Hand-crafting difference functions is hard
 - ▶ Not substantially different anomalies often well reconstructed
 2. **Discriminative Methods** on (deep) features (e.g. CNNs) distribution deviation
 - ▶ These often suffer from unsharp segmentation maps
 - ▶ Overfit to synthetic anomalies used during training and do not generalize well



¹ZKS21.

Framework Overview

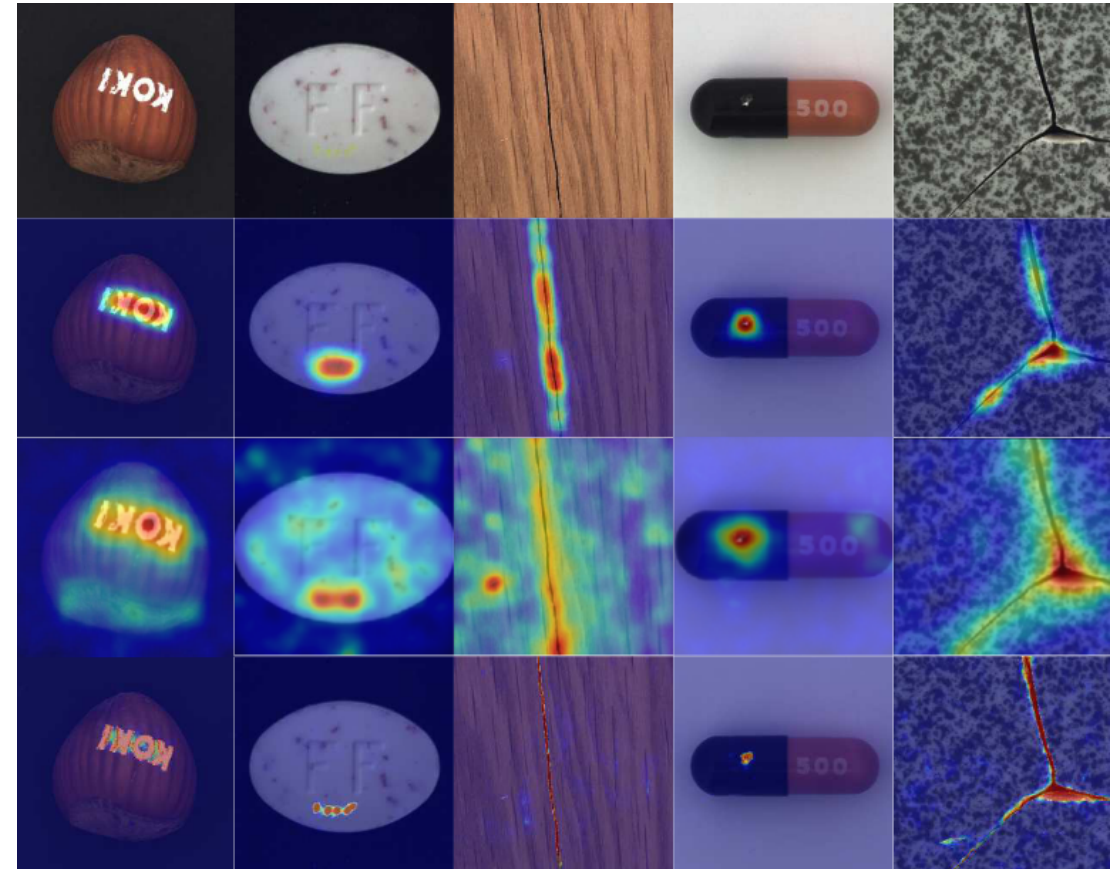
How it Works



Key Findings & Application & Conclusion

The Value of DRÆM

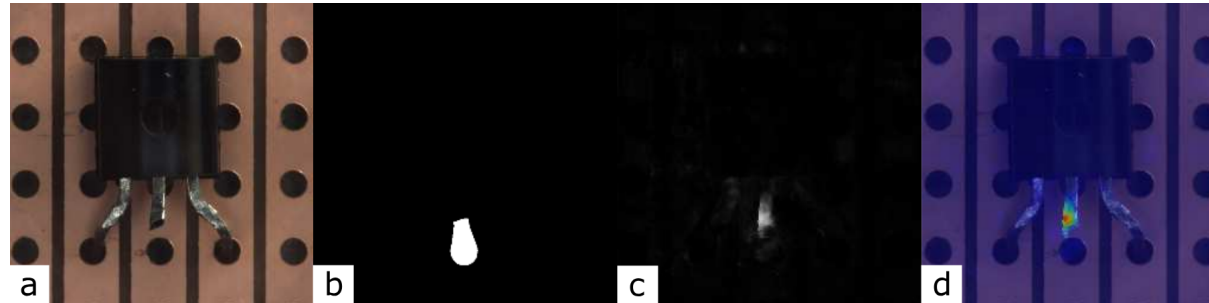
- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High quality segmentation can help downstream tasks
 - Estimate the size of impacted area
 - Distinguish between crack, corrosion or just paint
- We don't need expensive hand-labeled data
- We only need just-out-of-distribution patterns for anomaly generation



Limitation & Extension

Where to go from Here

- The method lacks performance in cases where domain knowledge is necessary to perform well
- For example here the whole lead of the transistor is cut off, but it only marks the position of the cut



- Another example could be when certain "normalities" are underrepresented in the training data
 - DRÆM may detect stains on walls as anomalies, while only structural damage should be detected
 - In these cases finetuning the discriminative network may improve performance

Thank you for your attention

The background of the slide is a solid blue color. In the lower half, there is a series of concentric, curved lines that create a sense of motion or a ripple effect, starting from the bottom center and curving outwards towards the sides.